2022A F=ma Exam: Problem 23

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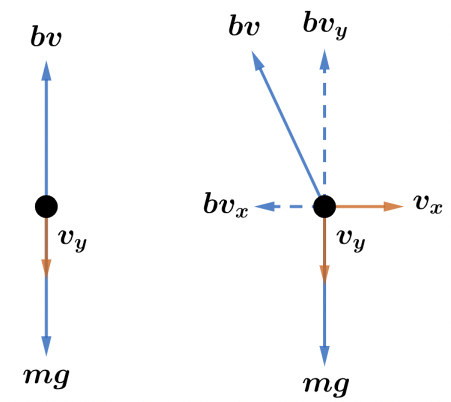
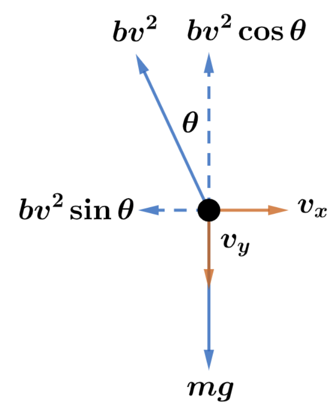


For linear drag, the vertical component of the drag force is bv_y and the horizontal component of the drag force is bv_x i.e. the motions in the vertical and horizontal directions are decoupled from each other. Thus, $t_A = t_B$.



For quadratic drag, the vertical component of the drag force is

$$F_y = bv^2 \cos\theta = bv^2 \left(\frac{v_y}{v}\right) = bvv_y = bv_y\sqrt{v_x^2 + v_y^2}$$

which is minimal when the object falls straight down, $v_x = 0$. Thus, $t_A > t_B$.

Hence, the answer is $\boxed{\text{C}}$.